

Network Forensics

Network Forensics

- the capture, recording, and analysis of network traffic to investigate security incidents or legal evidence
- ... also, the consequence of actions on digital networks
 - to identify suspicious behavior such as unauthorized access or data exfiltration
- encryption and obfuscation techniques make deep-packet analysis harder
- Tools
 - PCAP to capture network packets to analyze communication
 - (host.level) logs to investigate network device logs to reconstruct malicious activity
 - *Wireshark* for packet analysis, *Nmap* for port scanning, *Xplico* for application layer analysis, *NetworkMiner* for file extraction,

Evidence identification

- **digital footprint and online behavior of individuals or organizations involved in the investigation**
 - identifies potential sources of evidence
 - n social media profiles, email addresses, domain names, IP addresses, public repositories....
 - locate specific evidences
 - n public accounts related to the suspect or victim, affiliations, ...
 - reveals related accounts, aliases, or associated online identities that may not have been immediately apparent

Evidence collection

- **gather all relevant information without altering it**
- **OSINT may be used as a supplement to digital forensic data**
 - data from publicly available sources
 - n archived website snapshots (*Wayback Machine*)
 - n online information on registered domains
 - gather digital communications or posts by the suspect on public forums or social media
 - gathers metadata, IP history, or timestamps from online resources, which can support establishing timelines or verifying activities

Data Preservation

- **requirement: all evidence is preserved in its original state, ready for potential court use**
 - screenshots, web archives, metadata retrieval and all publicly available information
 - n adoption of tools that can generate evidence snapshots, including metadata, that allow future verification and defensibility of evidence in court

Analysis

- **support in examining the relationships, patterns, potential evidence connections, timeline reconstruction**
 - reveal connections between individuals, domains, IP addresses, or entities based on shared activities, posts, or affiliations
 - social network analysis, to help identification of assets, third parties involved in the case.
 - trace the origins of digital artifacts
 - emails or messages
 - identify the sources
 - support analysis of attack vectors and potential leak points
- origin of digital incidents (through correlation of username, profiles, IPs and known hacker groups)

Report

- **enrich findings to generate a comprehensive report**
 - incorporate attributable public information that helps to reinforce forensic findings
 - incorporates OSINT reports or visualizations to make connections and patterns clearer to readers.
 - e.g. network maps, timelines
 - OSINT sources allow for verification of online evidence, making the final report thorough and defensible.

Network Forensics – OSINT

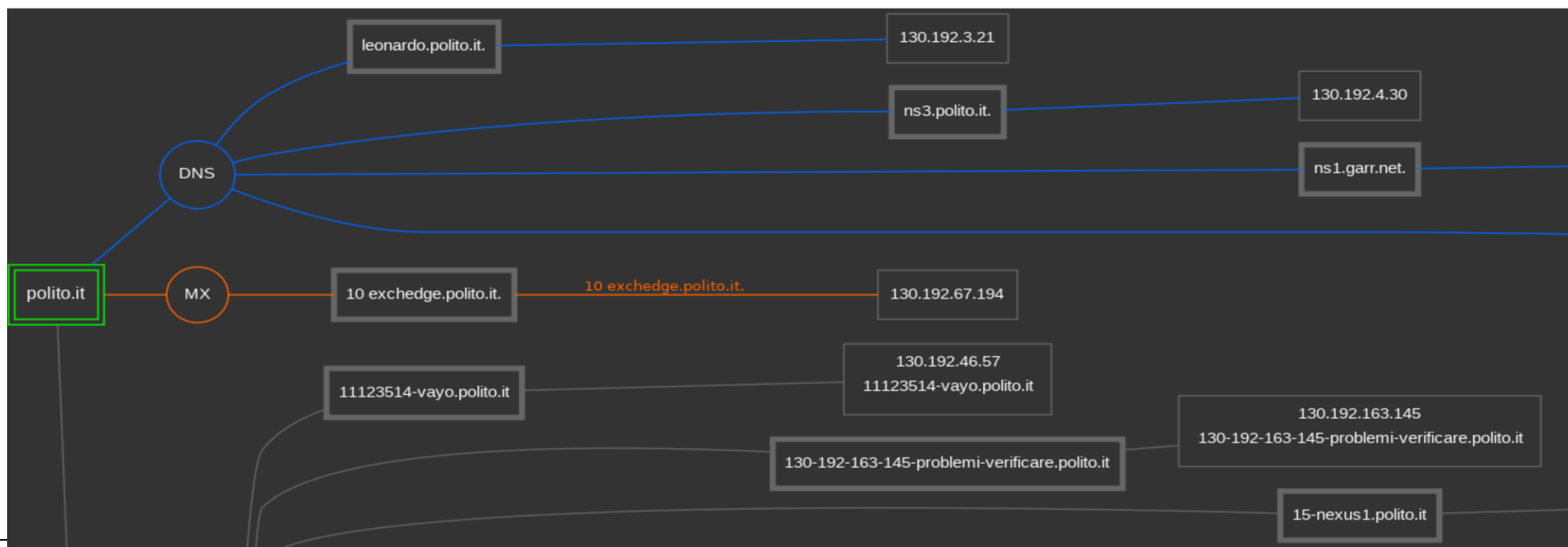
- contextual information about network components
 - Ip addresses
 - Narrow down the geographical location of user/servers
 - *e.g. IPinfo, ip-api.com, MaxMind GeoIP*
 - `ip:"8.8.4.4", hostname:"dns.google", anycast:true, city:"Mountain View", region:"California", country:"US", loc:"37.4056,-122.0775", org:"AS15169 Google LLC", postal:"94043", timezone:"America/Los_Angeles",`
 - `asn:Object, asn:"AS15169", name:"Google LLC", domain:"google.com", route:"8.8.4.0/24", type:"hosting",`
 - `company:Object, name:"Google LLC", domain:"google.com", type:"hosting"`
 - `privacy:Object, vpn:false, proxy:false, tor:false, relay:false, hosting:true, service:"",`
 - `abuse:Object, address:"US, CA, Mountain View, 1600 Amphitheatre Parkway, 94043", country:"US", email:"network-abuse@google.com", name:"Abuse", network:"8.8.4.0/24", phone:"+1-650-253-0000",`

Network Forensics – OSINT (II)

- Internet service provider and organizational data
 - Whois lookup
 - ownership and assignment details of IP addresses
 - e.g. RIPE NCC <https://apps.db.ripe.net/db-web-ui/query>
 - route: [130.192.0.0/16](#)
 - descr: TORINO-IT-LAN
 - origin: AS137
 - remarks: Politecnico di Torino
 - remarks: Università degli Studi di Torino
 - remarks: To notify abuse mailto: cert@garr.it
 - remarks: Send SPAM complaints to: spam.report@polito.it
 - mnt-by: [GARR-LIR](#)
 - created: 2002-04-24T11:36:29Z
 - last-modified: 2024-02-23T15:47:31Z
 - source: RIPE

Network Forensics – OSINT (III)

- domain exploration
 - reverse DNS lookups to identify domains associated with IP
 - can reveal related websites, services, subdomains (... and possible different purposes)
 - e.g. *DNSDumpster*, *Shodan*



Network Forensics – OSINT (IV)

- service detection
 - open ports and running services
 - e.g. *Nmap*
 - Databases of scanned IPs, revealing device type, versions, configurations
 - e.g. *Censys*, *Shodan*, *spiderfoot*

Network Forensics – OSINT (V)

- Mentions
 - IP addresses and network object In social media, paste sites
 - As part of security alerts and incidents
 - e.g. *Pastebin, TweetDeck ((obsolete)*
 - *site:pastebin.com "polimi.it"*
 - *site:pastebin.com "password" "polimi.it"*

Network Forensics – OSINT (VI)

- Threat intelligence feed
 - threat intelligence databases often mark suspicious IPs linked to malicious activity
 - e.g. *virusTotal*, *alienvault OTX*, ...

MALWARE FAMILIES: #LowFi:BRUTE:Win32/Iminent, SSH Brute-Force

ENDPOINT SECURITY Scan your endpoints for IOCs from this Pulse!

Indicators of Compromise (171K)

Related Pulses (0)

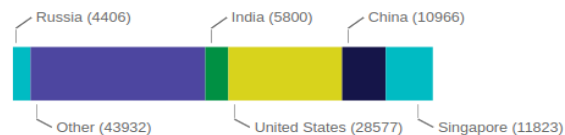
Comments (2)

History (3)

IPv4 (171350)



TYPES OF INDICATORS



THREAT INFRASTRUCTURE

Show 10 entries

TYPE	INDICATOR	ROLE	TITLE	ADDED
IPv4	122.191.109.66	bruteforce	SSH intrusion attempt from 122.191.109.66	Aug 11, 2024, 12:30:23 AM
IPv4	47.236.232.63	bruteforce	SSH intrusion attempt from 47.236.232.63	Aug 11, 2024, 1:05:59 AM
IPv4	1.13.181.190	bruteforce	SSH intrusion attempt from 1.13.181.190	Aug 11, 2024, 1:07:16 AM
IPv4	165.154.11.113	bruteforce	SSH intrusion attempt from 165.154.11.113	Aug 11, 2024, 1:16:23 AM
IPv4	190.183.61.91	bruteforce	SSH intrusion attempt from 190.183.61.91	Aug 11, 2024, 1:27:17 AM

Network Forensics – OSINT (VII)

- IP history
 - Previous association with different (suspicious) domains
 - e.g. *viewDNS.info*, *RiskIQ*

DNS Report for *polito.it*

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Parent Nameserver Tests

Status	Test Case	Information
	NS records listed at parent servers	Nameserver records returned by the parent servers are: giove.polito.it. [130.192.3.24] [TTL=3600] ns1.garr.net. [NO GLUE] [TTL=3600] ns3.polito.it. [130.192.4.30] [TTL=3600] leonardo.polito.it. [130.192.3.21] [TTL=3600] This information was kindly provided by a.dns.it.
	Domain listed at parent servers	Good! The parent servers have information on your domain. Some other domains (like .co.us) do not have a DNS zone at the parent servers.
	NS records listed at parent servers	Good! The parent servers have your NS records listed. If they didn't, people wouldn't be able to find your domain!
	Parent servers return glue	Good! The TLD of your domain (it) matches the TLD of your nameservers (it) and hence the parent servers MUST return the IP (glue) for your NS records... AND THEY DO!
	A record for each NS at parent	Oops! The parent servers don't have A records for each of your nameservers! This isn't a fatal error but means an extra lookup needs to be performed increasing the load time to your site.

Social Media Forensics

- **analysis of social media accounts and activities**
 - gathering information from public profiles, posts, and metadata
 - extract data from social media accounts
 - behavior and timing of activities
 - identify associations (e.g. linked accounts)

Social Media Forensics (II)

- **user profiling and attribution**
 - accounts cross-correlation
 - online activity metadata analysis
 - network graphing of interaction
 - e.g. *Maltego*, *spiderfoot*
- **geospatial analysis**
 - analysis of embedded data location (GPS, Exif data)
 - exiftool
 - reverse image search

Social Media Forensics (III)

- **relationships**
 - follower/following allow for identification of mutual friends
 - comments shows recurring interactions
- **content analysis**
 - trend and topics
 - user emotional state and motivation
- **image and video analysis**
 - reverse search (origin verification)
 - facial recognition

Social Media Forensics (IV)

- **monitoring of relevant activity**
 - hashtags, keyword
- **detection of fake information**
 - puppet profiles
 - analysis of fake activity
 - impersonating users
 - behavioral similarities across different profiles

Anti-forensics for network forensics (I)

- **obscure, alter, or remove traces of malicious activity on a network**
 - goal: evading detection and analysis by forensic investigators
- **obscure the truth of network flows**
- **hide intent and payload alike**
- **wipe or altered the true logs**
- **establish decoys and false flags**

Anti-forensics for network forensics (II)

■ encryption

- goal: prevent network content and traffic analysis
- concept: obscures the content of data packets, making it difficult to discern malicious intent or extract useful information.
- example: HTTPS, VPNs, IPSec

■ tunneling

- goal: prevent identification of illegitimate protocols/communication making hard to distinguish malicious activity from normal network behavior
- concept: encapsulating malicious traffic within legitimate protocols, such as HTTP or DNS.
- example: covert communication using DNS tunneling to exfiltrate data

Anti-forensics for network forensics (III)

- **packet manipulation**

- goal: misleads investigators and disrupts traffic reconstruction efforts
- concept: crafting or modifying packets to hide the origin, destination, or payload
- example: Spoofing IP addresses or using fragmented packets.

- **anti-traffic analysis techniques**

- goal: obfuscating patterns in network behavior to evade detection.
- concept: conceals malicious activity within a high volume of legitimate traffic
- example: noise generation (sending benign packets to blend malicious traffic with legitimate traffic)

Anti-forensics for network forensics (IV)

■ **time-based evasion**

- goal: avoids detection by analysts and automated systems
- concept: performing malicious activities at off-peak hours or in intervals to evade real-time monitoring.
- example: scheduled attacks or slow, incremental data exfiltration

○ **decoy traffic**

- goal: wastes resources and time during investigation
- concept: flooding the network with fake or irrelevant data to distract investigators
- example: botnets generating high volumes of benign traffic alongside malicious activity

Anti-forensics for network forensics (V)

■ **polymorphic and metamorphic malware**

- goal: bypasses IDS/IPS systems and other pattern-matching tools
- concepts: altering malicious payloads (changing encryption or rewriting the code) to avoid detection
- example: Wannacry that changes its payload with every packet transmission to bypass firewalls and IDS

■ **anonymization techniques**

- goal: masks the source of malicious activity, complicating attribution efforts
- concept: exploit tools/nodes/protocols to hide the identity and location of attackers
- example: proxies and/or TOR (The Onion Router)

Countering Anti-forensics

- **deep packet inspection (DPI)**
 - analyzing payloads and headers at the application level to uncover obfuscated content
- **encrypted traffic analysis**
 - using inspection tools/decrypting traffic (with authorization)
- **behavioral analytics**
 - to detect anomalies in network behavior irrespective of content (e.g. leveraging AI)
- **comprehensive logging**
 - implementing tamper-proof logging mechanisms like write-once-read-many (WORM) storage
- **threat intelligence integration**
 - databases of known malicious threats behaviors

Malware and endpoint forensics – OSINT (I)

- **Threat intelligent Feeds**

- how malwares operate
 - origins
 - behavior in the wild,
 - other instances of similar malware
 - track patterns of usage across campaigns
- Indicator of compromise
 - hashes
 - URLs/domains
 - file paths
 - registry keys

Malware and endpoint forensics – OSINT (II)

- **malware family**
 - common characteristics and strategies
 - allow for understanding of spread, impact, behavior, mitigation strategy
- **ongoing campaigns**
 - allow for threat groups identification
 - allow for prediction/identification of malware activity
 - allow for defense improvement
 - allow for code similarity
 - reuse of modules/techniques
- **gather information on C2 infrastructure**